# Guillermo Fernando Farfán Romero

## Senior Python Backend Engineer & Technical Lead

Salta, Argentina (UTC-3 · Open to Remote / Relocation) | +54 9 11 2187-1473 | fernando.farfan16@gmail.com

Portfolio: gfarfan.dev | LinkedIn | GitHub | Credly

## PROFESSIONAL SUMMARY

Senior Python Backend Engineer and Technical Lead with 8+ years of experience designing scalable microservices architectures, high-throughput RESTful APIs, and enterprise integration backends. Specialist in **Python 3.10+ (FastAPI, Asyncio, uvloop, Celery, SQLAlchemy 2.0, Pydantic v2)**, optimized data persistence in **PostgreSQL (with pgvector for semantic search and AI/LLM integrations) and Redis**, and containerized cloud workloads on **Google Cloud Platform (Cloud Run, GKE)**.

Strong track record leading the adoption of Clean Architecture, Domain-Driven Design (DDD), automated testing suites with Pytest under TDD methodologies, and distributed observability with **OpenTelemetry / IBM Instana**. Bridges business objectives and technical scalability with strategic team leadership, backed by a background in Information Systems and advanced coursework in Labor Relations (UBA).

## TECHNICAL SKILLS & COMPETENCIES

- **Languages & Backend Core:** Python 3.10+ (Asyncio, Strict Type Hints, Concurrency, Generators, uvloop), TypeScript / Node.js, Bash Scripting.
- **Python Frameworks & Ecosystem:** FastAPI, Pydantic v2, SQLAlchemy 2.0 (Async Engine & ORM), GenAI / LLM Integration (Vertex AI, OpenAI API, RAG, pgvector), Alembic (Zero-Downtime Migrations), Celery, Django / DRF, Flask, httpx, Uvicorn, Gunicorn, Streaming APIs (SSE / WebSockets).
- **Databases, Caching & Messaging:** PostgreSQL (Advanced Modeling, pgvector, Composite Indexes, JSONB, asyncpg, PgBouncer), Redis (Distributed Caching, Rate Limiting, Pub-Sub, Celery Broker), MySQL, MongoDB, Google Cloud Pub/Sub.
- **Cloud & DevOps:** Google Cloud Platform (Cloud Run, GKE, Cloud Storage, Secret Manager), Docker (Multi-stage builds), Docker Compose, Terraform, GitHub Actions CI/CD.
- **Architecture & Engineering Quality:** Microservices, Clean Architecture, Hexagonal Architecture (Ports & Adapters), Domain-Driven Design (DDD), Event-Driven Architecture, Architecture Decision Records (ADRs), REST APIs, OpenAPI/Swagger, TDD/BDD (Pytest, Factory Boy).
- **Observability & Reliability:** OpenTelemetry (OTel), IBM Instana APM, Zabbix, Sentry, Prometheus, Git, Postman, SonarQube.

## PROFESSIONAL EXPERIENCE

### Technical Lead & Senior Systems Engineer (Python Backend Focus) | Reforest Latam

*May 2026 – Present | Full-time (Remote)*

- **Led the adoption of Clean Architecture and Domain-Driven Design (DDD)** across an engineering team of 6 for environmental telemetry platforms, formalizing Architecture Decision Records (ADRs) and **accelerating modular feature delivery by 35%**.
- **Established >85% automated test coverage across microservices** by architecting robust test suites with **Pytest** under TDD practices.
- **Orchestrated distributed asynchronous workers with Celery and Redis**, integrated with asynchronous **FastAPI** endpoints and **Cloud Pub/Sub** for decoupled, high-scale background event processing.
- **Engineered cloud-native security and integration layers** connecting microservices to GCP resources (Cloud Storage, Secret Manager) under JWT/OAuth2 authentication and least-privilege principles.
- **Implemented automated CI/CD workflows in GitHub Actions**, enabling continuous **zero-downtime deployments** to **Cloud Run** and GKE.

### Senior Python & Software Consultant | Independent Consulting & Enterprise Projects

*January 2021 – Present | Strategic B2B Technology Consultant (Independent / Contract / Remote)*

- **Eliminated N+1 query bottlenecks and optimized asynchronous persistence with SQLAlchemy 2.0** (selective *eager loading*), strict modeling with **Pydantic v2**, and caching with **Redis**, driving a **>60% reduction in p95 latency on critical endpoints**.
- **Architected and built ingestion microservices and transactional sync engines** connecting enterprise ERP/SAP platforms with relational databases for international retail, mining, and healthcare corporations.
- **Designed high-concurrency, mission-critical REST APIs in FastAPI**, guaranteeing OpenAPI/Swagger contracts, rigorous schema validation, asynchronous streaming, and structured exception handling.
- **Standardized reproducible development and production environments** through multi-stage Docker builds and automated infrastructure provisioning with **Terraform** on Google Cloud.

**Senior Software Engineer & Database Administrator | University of Buenos Aires (UBA)**
*March 2018 – February 2024 | Institutional Systems Specialist (Part-time / Buenos Aires, Argentina)*

- **Engineered concurrent asynchronous Python microservices and ETL data synchronization pipelines** (+50,000 users), **slashing batch processing runtimes from hours to minutes**.
- **Implemented database schema versioning and governance with Alembic**, guaranteeing transactional data integrity and automated testing suites on PostgreSQL and SQL Server.
- **Optimized complex SQL queries and stored procedures**, removing computation bottlenecks across high-demand transactional database servers.

**Systems Infrastructure & Operations Coordinator | Government of the City of Buenos Aires (GCBA)**
*January 2020 – December 2023 | Full-time (Buenos Aires, Argentina)*

- **Established technical interoperability standards and automated validation layers in backend services**, achieving a **40% reduction in cross-departmental data inconsistencies**.
- **Maintained a 99.9% operational uptime SLA** across municipal platforms through code reviews and strict software release governance.

## EDUCATION

- **Associate Degree in Information Systems** | National Technological University (UTN) *(In Progress — Expected Graduation: December 2027)*
- **Bachelor's Studies in Labor Relations & Human Resources** | University of Buenos Aires (UBA) *(Advanced University Coursework Completed — Focus on Organizational Leadership, Talent Strategy & Engineering Team Dynamics)*

## KEY CERTIFICATIONS *(100% VERIFIABLE ON CREDLY)*

- **Backend & Java Certifications (Credly):** Meta Back-End Developer Certificate | IBM Java Developer Professional Certificate V2 | IBM JavaScript Backend Developer Professional Certificate V2.
- **DevOps & Software Engineering Certifications (Credly):** IBM DevOps and Software Engineering Professional Certificate | IBM Full Stack Software Developer Certificate | Meta Full-Stack Engineer Certificate.

## LANGUAGES

- **Spanish:** Native
- **English:** Professional Working Proficiency (B2 — Fluent technical communication for live engineering syncs, RFCs, technical documentation, and cross-functional collaboration)